

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 2

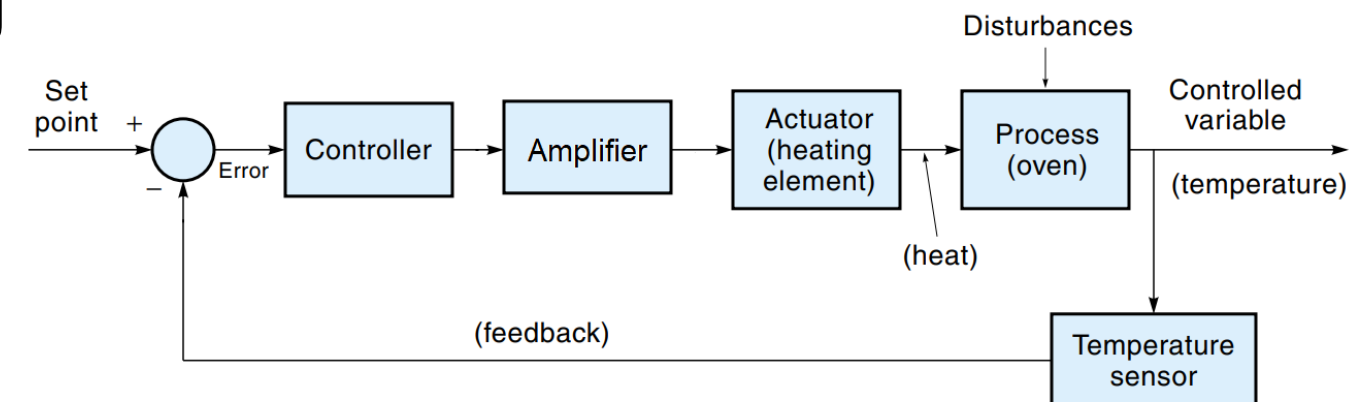
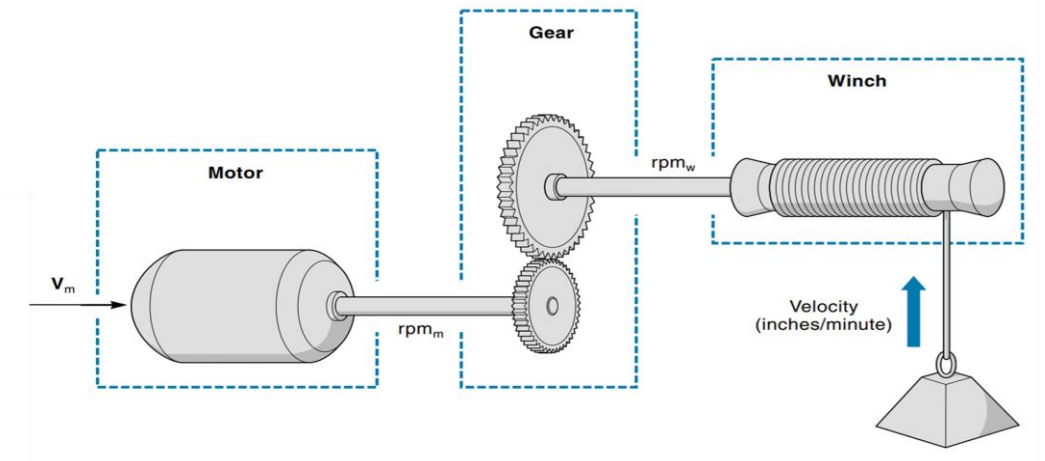
LECTURE 2

Translational Mechanical Systems

- Modeling of Translational Mechanical Systems
- System Decomposition & Model Complexity
- Variables & Elements
- Element Laws
- Interconnection Laws
- Equation of Motion

What We Already Know?

- A **dynamic system** is a collection of **components** and **circuits** connected together to perform a useful function.
- Each element in the system **converts energy** from one form to another.
- In order to analysis, design and control of dynamic systems, we need to first understand the **input-output relationship** of the system elements and find a **model** for them.
- The model helps us to figure out:
 - How components affect each other and the overall system.
 - How the output of the system will react to different inputs.
 - How to analyze, predict and control the system behavior.
- **System modeling** means finding a **mathematical equation** describing how the **output** of the system is related to its **input**.



How to find the system model?

What We Already Know?

□ Lump-Parametric Modeling

- Find the input-output relationships for systems by considering them to be composed of just a few **simple basic elements** and applying the **physical laws** from **first-principles**.
- Mainly used to model **electrical**, **mechanical**, and **electromechanical** systems.
- The model is obtained as a **differential equation**, then shown in any standard form of, **Transfer Function model**, **State-Space model** or **Block diagram model**.
 - **Electrical systems** can be considered as compose of basic elements, which can be represented by **resistors**, **capacitors**, **inductors** and **op-amps**.
 - These are characterized by voltage –current relationships for components and the laws of interconnection **Kirchhoff's Voltage Law (KVL)** and **Current Law (KCL)**
 - **Mechanical systems** can be considered as compose of basic elements, which can be represented by **springs**, **dampers** and **masses**, and are characterized by **Newton's Laws of motion**.

□ Empirical Modeling

- An **experimental approach**, which mainly used to model **thermal** and **fluid (hydraulic and pneumatic)** processes.
- Some **experiments** are performed on the system to **collect input-output data**, a model is then fitted to the collected data by assigning suitable numerical values to its parameters.

Modeling of Mechanical Systems

- The motion of elements of mechanical systems can be described as:
 - **Translational Motion**
 - **Rotational Motion**
- The equations governing the motion of mechanical systems are called the **equation of motion** that often directly or indirectly formulated by applying **Newton's law of motion** to the **free-body diagram** (FBD).

Translational Motion

$$\sum F_{ext} = Ma$$

$$\sum T_{ext} = J\alpha$$

Rotational Motion

- The number of equations of motion required is equal to the number of **linearly independent** motions or the number of **degrees of freedom**.
 - **Step 1:** Identify **reference point** and **positive direction** of motion.
 - **Step 2:** Draw a **free-body diagram** for each mass/point of motion.
 - **Step 3:** For each free-body diagram, find the forces acting on the body due only to its own motion and the forces create by the adjacent motion.
 - **Step 4:** Use **Newton's law** on each body to form the **differential equation of motion**.

Translational Mechanical Systems : Variables & Elements

- The **translational motion** is defined as a motion that takes place along a **straight** or **curved** path.
- The **variables** that are used to describe the translational motion are:
 - $f(t)$: Force (N)
 - $x(t)$: Displacement (m)
 - $v(t)$: Velocity (m/s)
 - $a(t)$: Acceleration (m/s²)
- All these variables are function of time.
- **Displacements** are measured with respect to reference condition, which is the equilibrium position of the body.
- **Velocities** and **accelerations** are normally expressed as the derivatives of the corresponding **displacement**.

$$v(t) = \frac{dx(t)}{dt} = x'(t) = \dot{x}(t)$$

$$a(t) = \frac{d^2x(t)}{dt^2} = x''(t) = \ddot{x}(t)$$

- The **elements** that we include in translational systems are:
 - Inertia Element → **Mass**
 - Stiffness Element → **Spring**
 - Damping Element → **Damper**

Translational Mechanical Systems: Example

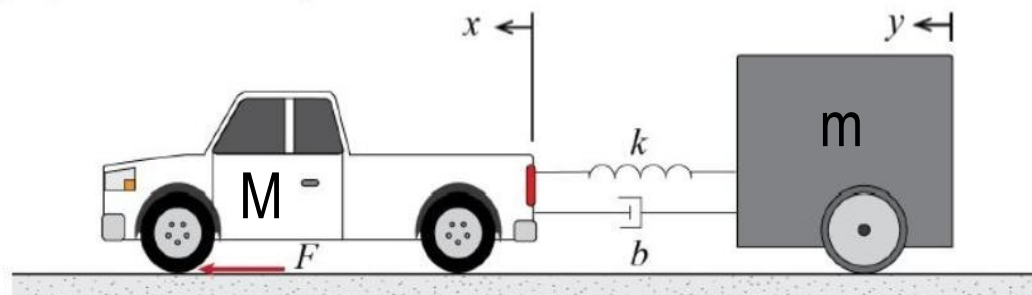
- Some real-world examples of translational motion systems.

- **Truck Pulling a Cart**



Input: Applied engine force

Outputs: Displacement of truck and cart

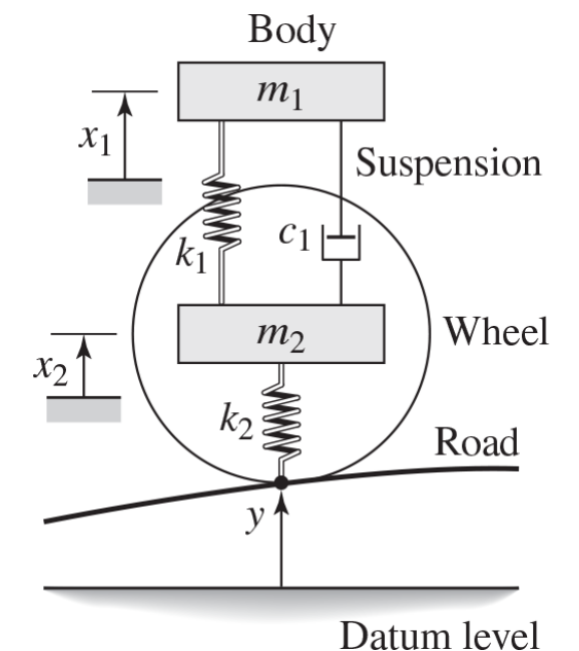


- **Vehicle Suspension System**



Input: Vertical displacement due to road bumps

Outputs: Vertical displacement of car



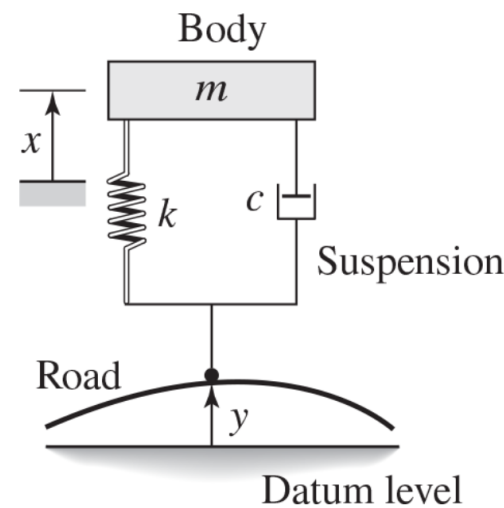
System Decomposition & Model Complexity

- The **complexity** of the model depends on its **intended use**.
- For example, consider the **vehicle suspension system**.
- The **quarter car suspension system** can be modeled as depicted in either model A or B.
- **Model A** is a simple model that includes the spring and damping due to the shock absorber, and mass of the front quarter of the vehicle.

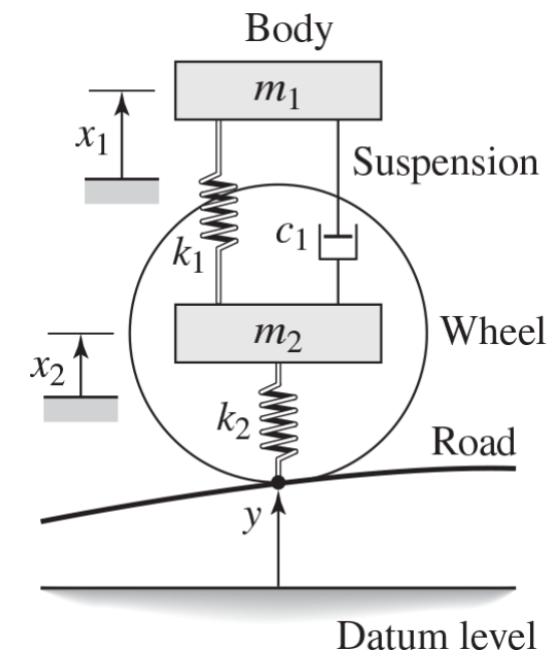
- This model is adequate for predicting the vertical motion of the front quarter for relatively **slow (low frequency) inputs**.

- An example might be a series of regularly spaced **speed bumps** the vehicle travels over at slow to moderate speeds as it might do in a **parking lot** or **residential area**.

- At such speeds, the **suspension** absorbs most of the energy and accounts for most of the dynamics.



Model A

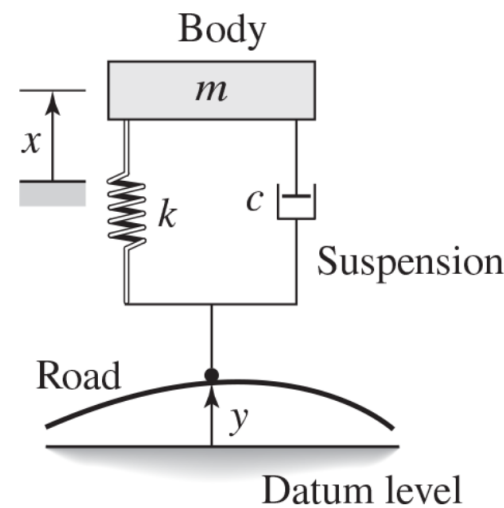


Model B

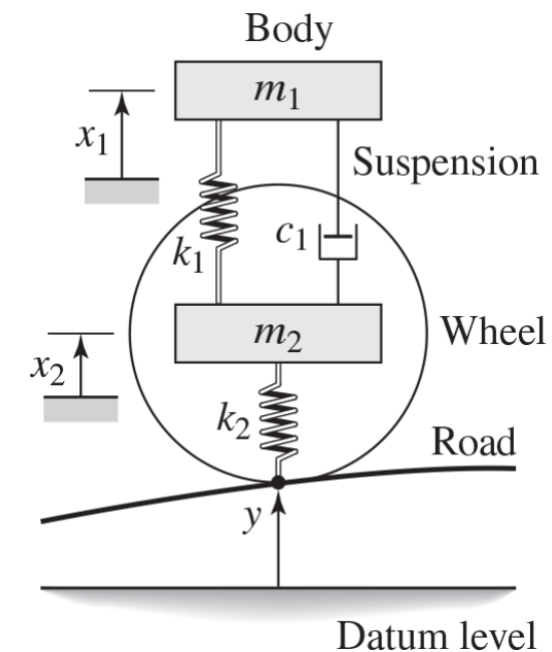


System Decomposition & Model Complexity

- The **complexity** of the model depends on its **intended use**.
- For example, consider the **vehicle suspension system**.
- The **quarter car suspension system** can be modeled as depicted in either model A or B.
- **Model B** is a more elaborate model that includes compliance of the tire due to the compressed air within, and the combined mass of the wheel, tire and brake assembly, as well.
 - This model can be used for predicting the vertical motion of the front quarter for **high frequency inputs**.
- For example, at highway speeds, the suspension receives **more frequent** vertical displacements with **small amplitudes** from the highway markers.
- At such frequencies, the vehicle suspension does not have sufficient time to respond and absorb the energy. Instead, the tires absorb much of the energy and play a more critical role in predicting the dynamic response and induced vibrations.



Model A



Model B

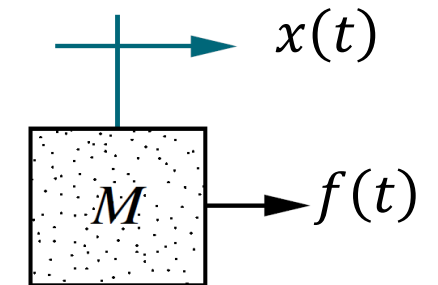


Translational Mechanical Systems: Element Laws

□ Inertia Element: Mass

- If a **force** is applied on a body, then it is opposed by an opposing force due to **inertia**.
- The inertia of a body can be represented by a **mass**.
- From the **Newton's second law**, the **applied force** $f(t)$ is proportional to the **acceleration** $a(t)$ of the body.
- The M is the **mass of the body**. The unit is (kg).

$$f(t) = Ma(t) = M \frac{dv(t)}{dt} = M \frac{d^2x(t)}{dt^2}$$



- **Energy** in a mass is stored as **kinetic energy** if the mass is in motion, and as **potential energy** if the mass has a vertical displacement relative to the reference point.

$$KE = \frac{1}{2} Mv^2$$

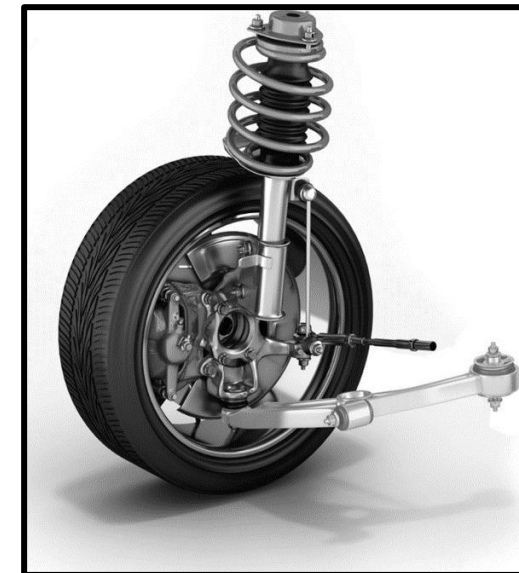
$$PE = Mgh$$

where $g = 9.81 \text{ m/s}^2$ is **gravitational acceleration** and h is the **height of mass above its reference position**.

Translational Mechanical Systems: Element Laws

□ Stiffness Element

- All physical objects **deform** somewhat under the action of externally applied **forces**.
- When the **deformation is negligible** for the purpose of the analysis, we can treat the object as a **rigid body**.
- Sometimes, an elastic element is intentionally included in the system, as with a **spring in a vehicle suspension**.
- Sometimes the element is not intended to be elastic but **deforms** anyway because it is subjected to large forces, such as **cables**, **beams** and **rods**.
- This can be the case with the **boom or cables of a large crane that lifts a heavy load**.
- In such cases, we must include the deformation and corresponding forces in our analysis.



Translational Mechanical Systems: Element Laws

□ Stiffness Element: Spring

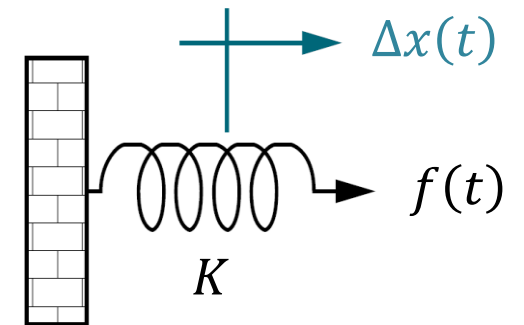
- Any mechanical element that undergoes a change in shape when subjected to a force can be characterized by a **stiffness element**, provided only that an **algebraic** relationship exists between the **elongation** and the **force**.
- The stiffness of an element can be represented by a **linear spring**.
- If a **force** is applied to a spring, then it is opposed by an opposing force due to the **elasticity** of the spring.
- From the **Hooke's law**, the **applied force** $f(t)$ to a spring is proportional to the **displacement** $\Delta x(t)$ of the spring.
- The K is the **spring constant**. The unit is (N/m).

$$f(t) = K\Delta x(t)$$

- Potential energy stored in a spring** that has been stretched or compressed, and for a linear spring that energy is given by:

$$PE = \frac{1}{2} K (\Delta x)^2$$

- An **ideal spring** element is massless.
- A **real spring** element can be represented by an ideal element either by neglecting its mass or by including it in another mass in the system.

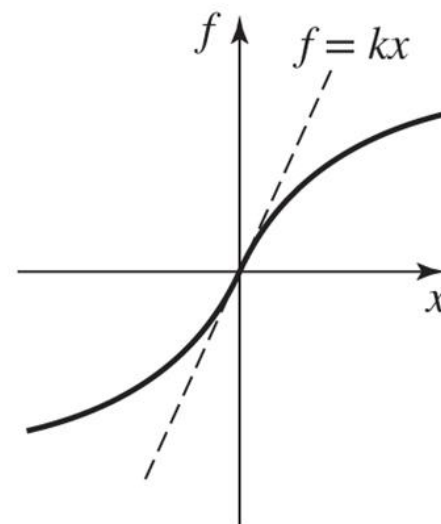
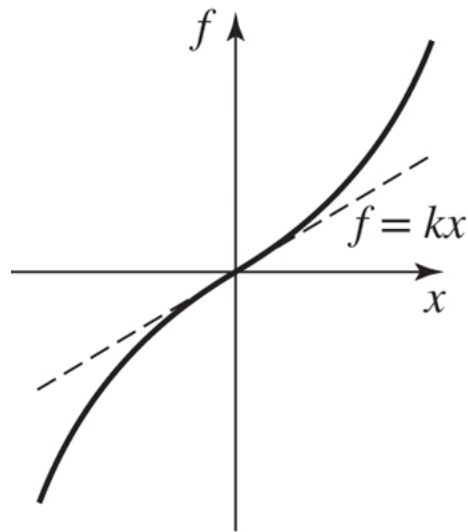


Translational Mechanical Systems: Element Laws

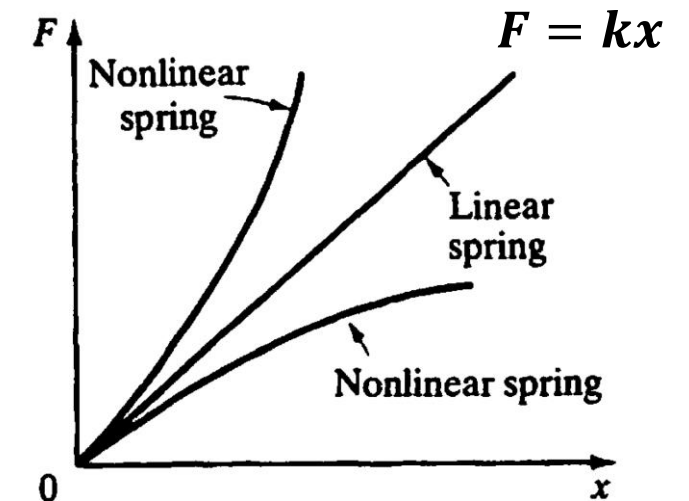
□ Properties of Spring

- For practical springs, the solution of linearity may be good only for **relatively small net displacements**.
- When a linear spring is stretched a point is reached in which the force per unit displacement begins to change and the spring becomes a **nonlinear spring**.
- **Spring constants** indicate **stiffness** of the spring:
 - **Large** value of spring constants correspond to a **hard spring**,
 - **Small** value of spring constants to a **soft spring**.

Nonlinear
characteristics of a
hardening spring



Nonlinear
characteristics of a
softening spring



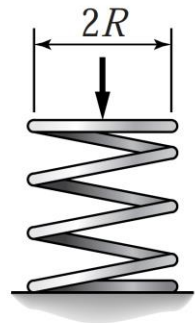
- Mechanical elements can be described by the **linear law** $f = kx$, for both **compression** and **extension**, as long as the deformations are not beyond the elastic limit.

Translational Mechanical Systems: Element Laws

□ Spring Constant

- The spring constant can be determined experimentally by a [tension test](#) or analytically from the [geometry and material properties](#).

Coil spring



$$K = \frac{Gd^4}{64nR^3}$$

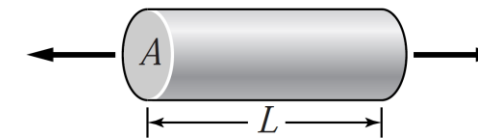
d = wire diameter

n = number of coils

R = radius of the coil

G = Shear modulus of elasticity

Solid rod



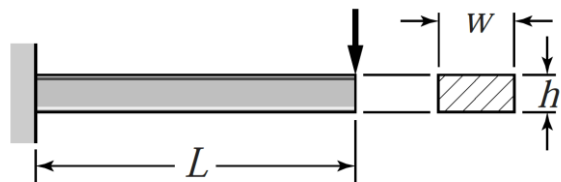
$$K = \frac{EA}{L}$$

A = rod area

L = rod length

E = Shear modulus of elasticity

Cantilever beam



$$K = \frac{Ewh^3}{4L^3}$$

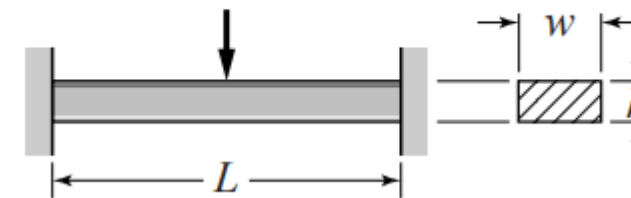
w = beam width

h = beam thickness

L = beam length

E = Shear modulus of elasticity

Fixed-end beam



$$K = \frac{16Ewh^3}{L^3}$$

w = beam width

h = beam thickness

L = beam length

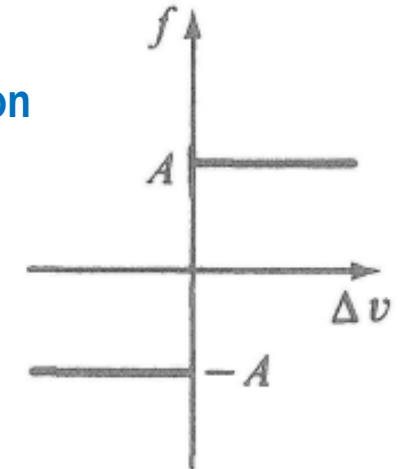
E = Shear modulus of elasticity

Translational Mechanical Systems: Element Laws

□ Damping Element

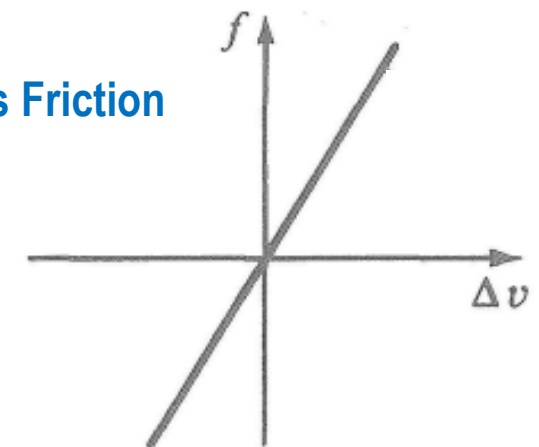
- A **damping element** represents the **friction** characteristics in the motion.
- There are two main type of friction forces:
 - **Coulomb Friction or Dry Friction**
 - A **friction force** caused by relative motion of two **solid surfaces** in contact.
 - It has two types: **Static** friction and **Kinetic** friction

Coulomb Friction



- **Viscous Friction**
 - A **fluid resistance force** produced by a **fluid layer** moving relative to a **solid surface**.
 - For example, a solid mass sliding on an oil film that has laminar flow is subject to viscous friction
 - If the fluid flow is laminar, the relationship between **viscous friction** and the **relative velocity** can be represented as a **linear** function.

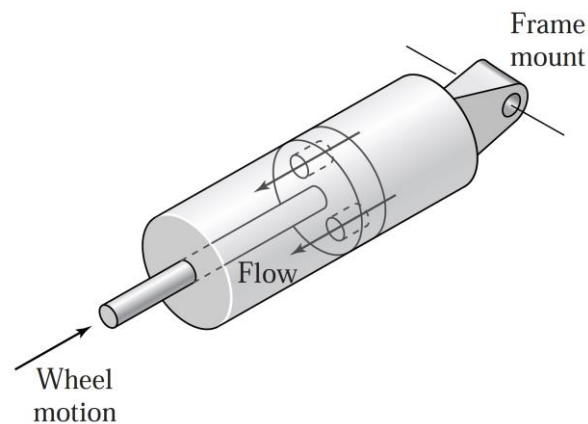
Viscous Friction



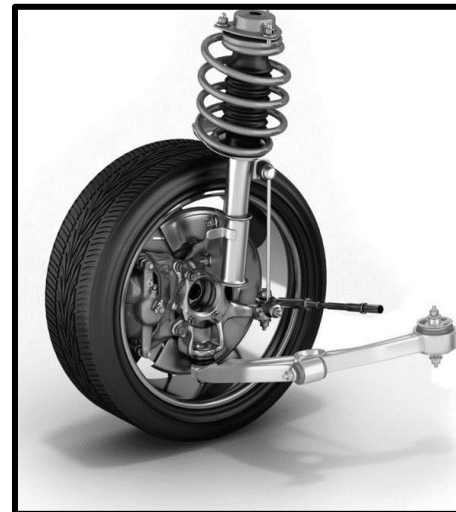
Translational Mechanical Systems: Element Laws

□ Damping Element

- Engineering systems can exhibit viscous damping in **bearings** and other **surfaces lubricated** to prevent wear.
- Sometimes, damping elements can be deliberately included as part of the design.
- For instance, a **dashpot** or a **shock absorber** in a **door closer** or **in a vehicle suspension**, which consists of a **piston** and an **oil filled cylinder**.
- Any relative motion between the piston rod and the cylinder is resisted by oil because oil must flow around the piston from one side to the other.



Piston Damper



Shock Absorber



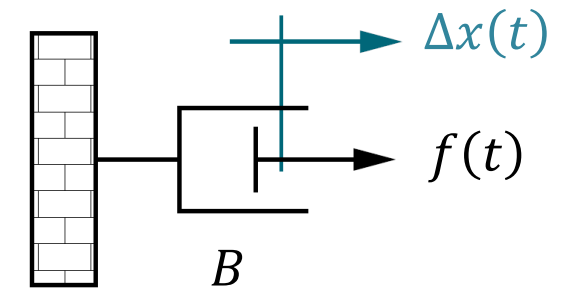
**Pneumatic
Door Closer**

Translational Mechanical Systems: Element Laws

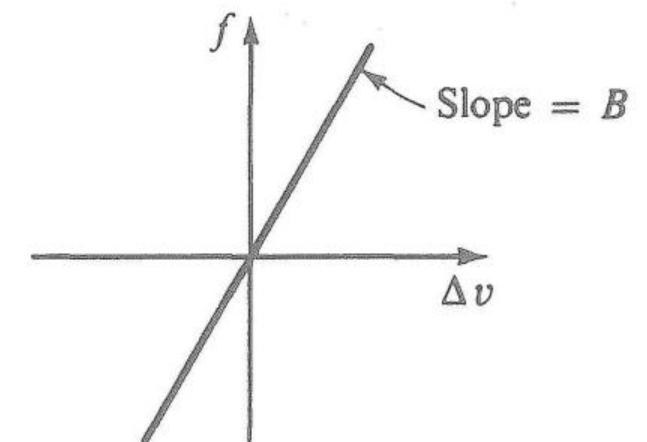
□ Damping Element: Damper

- A **damping element** is an element that resists **relative velocity** across it.
- The **damping effect** in a system can be represented by a **linear damper**.
- If a **force** is applied to a damper, then it is opposed by an opposing force due to **viscous friction** of the damper.
- The **applied force** $f(t)$ is proportional to the **velocity** $v(t)$ of the motion.
- The B is the **viscous friction coefficient**. The unit is (N.s/m)

$$f(t) = B\Delta v(t) = B \frac{d\Delta x(t)}{dt}$$



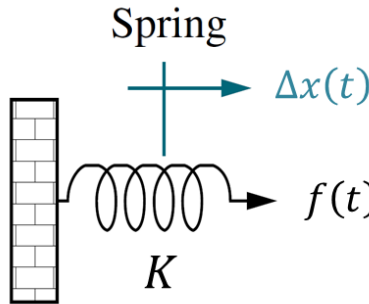
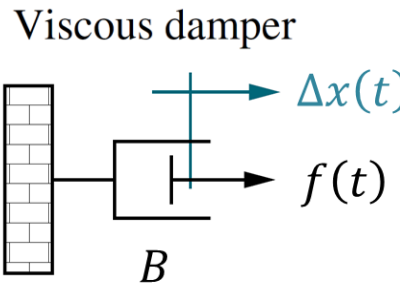
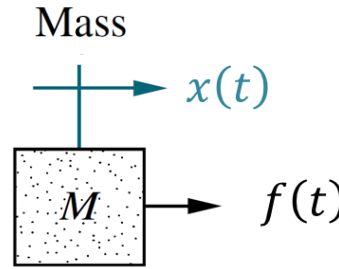
- The **viscous friction coefficient** B is proportional to the contact area and the viscosity of the fluid.
- Essentially, the damper **absorbs energy**, and the absorbed energy is **dissipated as heat** that flows away to the surroundings.



Translational Mechanical Systems: Element Laws

□ Summary

- Table shows summary of the Force-velocity, and Force-displacement translational relationships for Spring, Viscous dampers, and Mass.

Element	Force-velocity	Force-displacement
Spring 	$f(t) = K \int_0^t \Delta v(t) dt$	$f(t) = K \Delta x(t)$
Viscous damper 	$f(t) = B \Delta v(t)$	$f(t) = B \frac{d\Delta x(t)}{dt}$
Mass 	$f(t) = M \frac{dv(t)}{dt}$	$f(t) = M \frac{d^2 x(t)}{dt^2}$

Translational Mechanical Systems: Interconnection Laws

□ Newton's Second Law

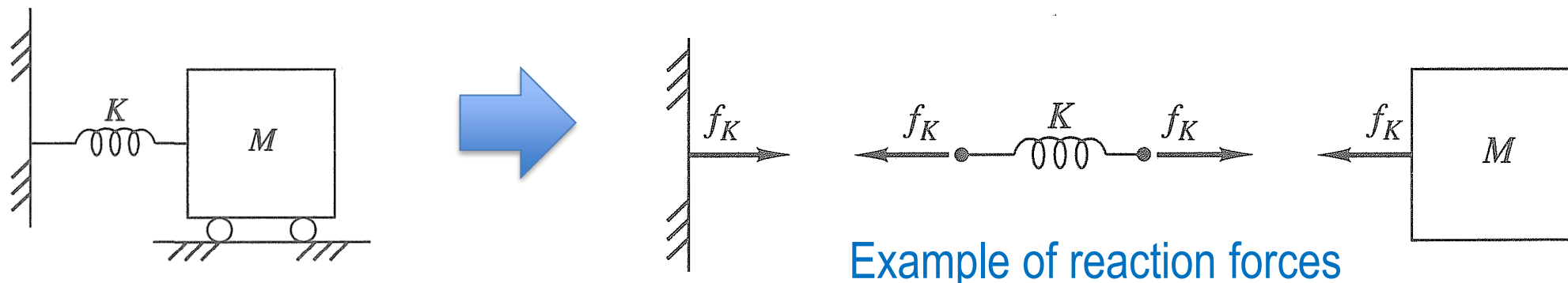
- Suppose that forces are acting on a body of **mass** m .
- If $\sum f$ is the **sum of all forces** acting on mass m through the center of mass in the given direction, then,

$$\sum f_{ext} = ma$$

where $a(t)$ is the resulting absolute **acceleration** in that direction.

□ Newton's Third Law: The Law of Reaction Forces

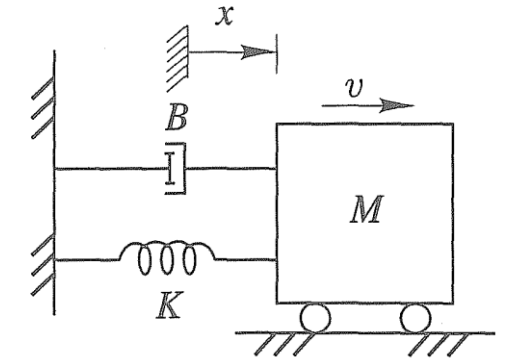
- In order to relate the forces exerted by the elements of **damping** and **stiffness** to the forces acting on a **mass** or **junction point**, we need **Newton's third law** regarding reaction forces.
- Accompanying any force of one element on another, there is a reaction force on the first element of equal magnitude and opposite direction.



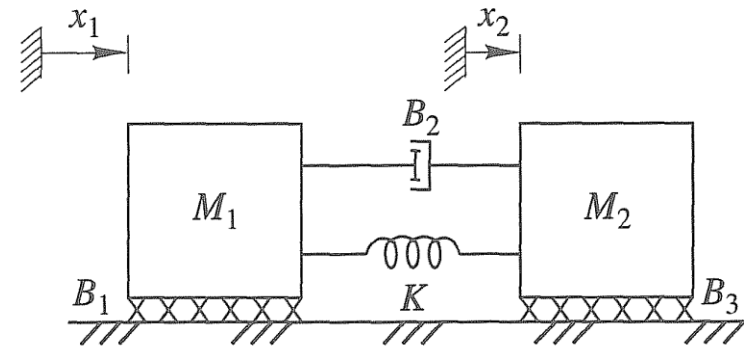
Translational Mechanical Systems: Interconnection Laws

□ The Law for Displacement

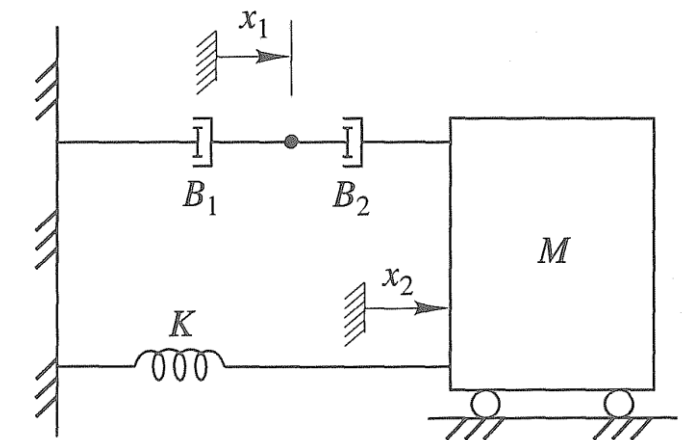
- If the ends of two elements are connected, those ends are forced to move with the **same displacement** and **velocity**.
- For example, because the damper and spring are both connected between the wall and the mass, the right ends of both elements have the same displacement x and move with the same velocity v .



- In this system, where B_2 and K are connected between two moving masses, the elongation of both elements is $x_2 - x_1$.



- Let x_1 and x_2 denote displacements measured with respect to reference position. Then the respective elongations of B_1 , B_2 and K are x_1 , $x_2 - x_1$ and x_2 .



Modeling of Translational Mechanical Systems

Example 1

Find the equation of motion for the given mass-spring-damper system. Assume that the input is the applied force, $f(t)$, and the output is the displacement of the mass, $x(t)$.

Assume that the positive direction for motion is to the right.

Draw the free-body diagram of the system for the mass M , showing just the external forces acting on the mass.

Apply Newton's second law to find the equation of motion.

$$\sum f_{ext} = ma \rightarrow f(t) - f_K(t) - f_B(t) = Ma(t)$$
$$f(t) - Kx(t) - B\dot{x}(t) = M\ddot{x}(t)$$

$$M\ddot{x}(t) + B\dot{x}(t) + Kx(t) = f(t)$$

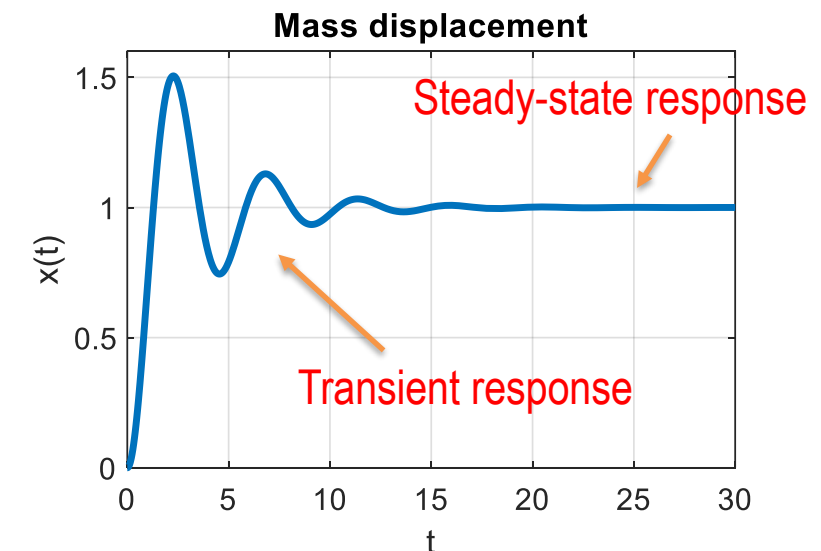
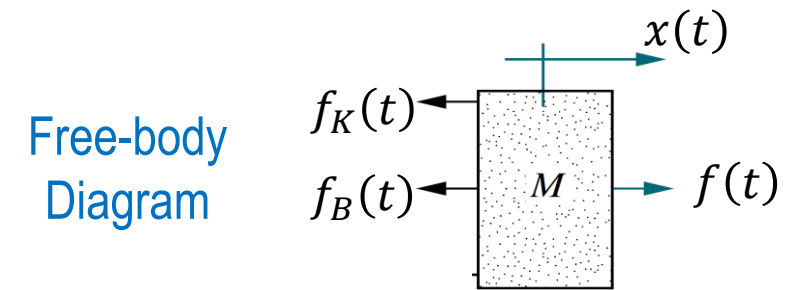
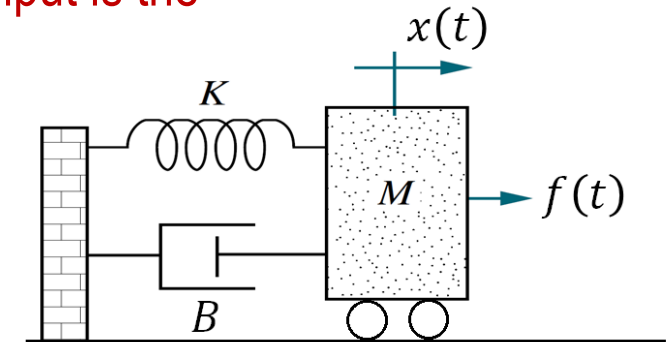
Second-order
Differential Equation

Assume $M = 50\text{kg}$, $B = 30\text{Ns/m}$ and $K = 100\text{N/m}$ we have:

$$50x''(t) + 30x'(t) + 100x(t) = f(t)$$

Assume applied force is $f(t) = 100\text{N}$, solution of the second-order differential equation is:

$$x(t) = 1 - e^{-0.3t}(0.217 \sin(1.38 t) + \cos(1.38 t))$$



Modeling of Translational Mechanical Systems

Example 2

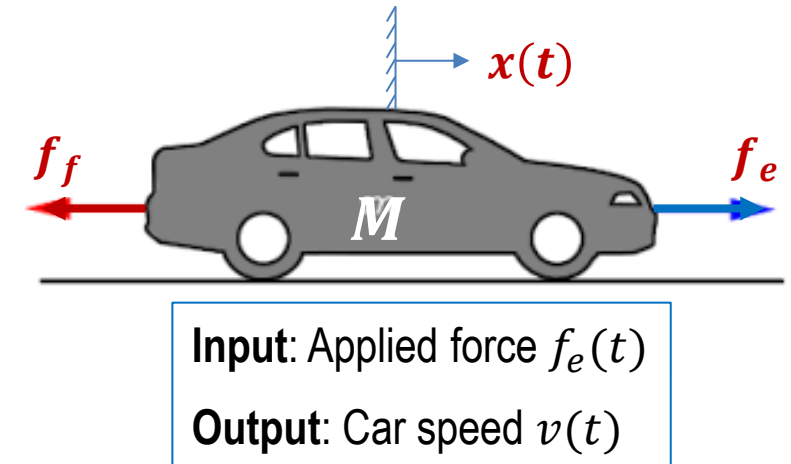
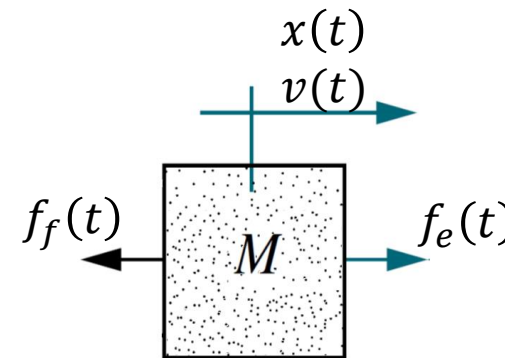
Consider a simple car cruise system. Assume that the engine applies a forward force of $f_e(t)$ and air friction is proportional to the car's speed $v(t)$. Find the equation to relate the applied force to the car's speed.

To obtain a system model we draw **free-body** diagrams, these being diagrams of masses showing just the **external forces** acting on each mass.

Apply **Newton's second law** as below

$$\sum f_{ext} = ma \rightarrow f_e(t) - f_f(t) = Ma(t)$$

$$f_e(t) - Bv(t) = Mv'(t)$$



The **differential equation** relating speed of the car $v(t)$ to the engine force $f_e(t)$ is determined as

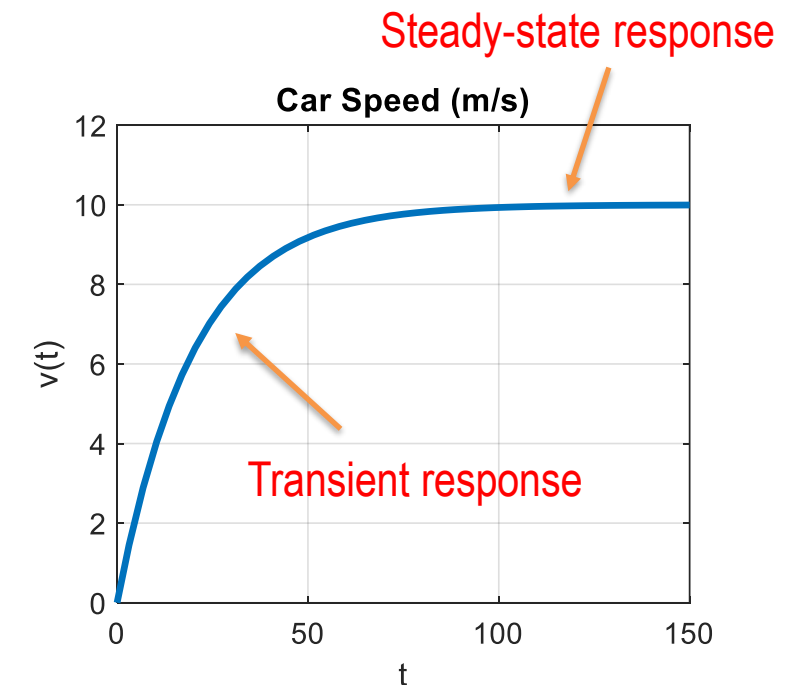
$$f_e(t) = Mv'(t) + Bv(t) \quad \text{First-order differential equation}$$

where M is the total mass and B is the viscous damping coefficient for air friction.

Assuming $M = 1000\text{kg}$ and $B = 50\text{Ns/m}$ we have: $f_e(t) = 1000v'(t) + 50v(t)$

If the applied force is $f(t) = 500\text{N}$, and initial speed is zero. The solution of the first-order differential equation is:

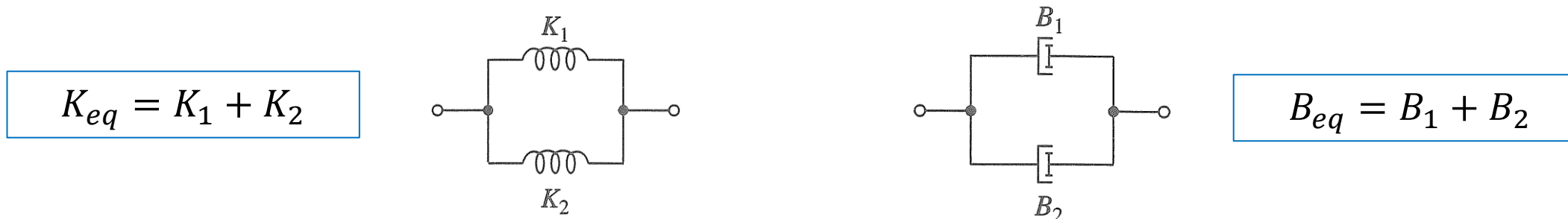
$$v(t) = 10(1 - e^{-t/20})$$



Modeling of Translational Mechanical Systems

□ Parallel & Series Combination

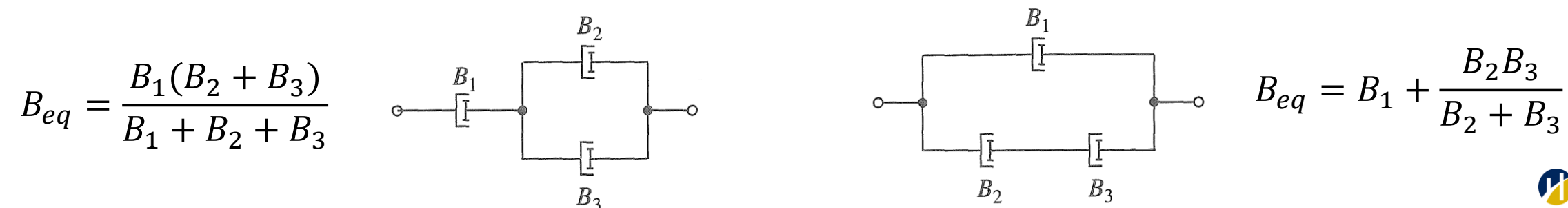
- In some cases, two or more **springs** or **dampers** can be replaced by a single equivalent element.
- Two **springs** or **dampers** are said to be **in parallel** if the first end of each is attached to the same body and if the remaining ends are also attached to a common body.
- The key requirement for parallel elements is that **respective ends move with the same displacement**.



- Two **springs** or **dampers** are said to be **in series** if they are joined at only one end of each element and if there is **no other element connected to their common junction**.



- It is also possible to have combination of **series and parallel connections**. For example:



Modeling of Translational Mechanical Systems

Example 3

Find the equation describing the motion of the mass in the translational system shown below. Show that the two springs can be replaced by a single equivalent spring, and the three friction elements by an equivalent element.

Assume that the positive direction for motion is to the right.

Draw the free-body diagram of the system for the mass M.

Apply Newton's second law to find the equation of motion.

$$\sum f_{ext} = ma$$

Mass $M \rightarrow f_a(t) - K_1x - B_1\dot{x} - B_2\dot{x} - B_3\dot{x} - K_2x = M\ddot{x}$

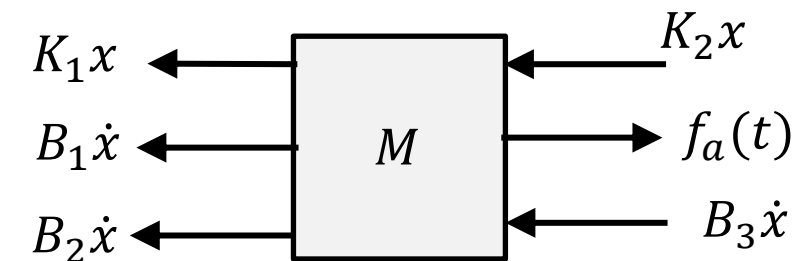
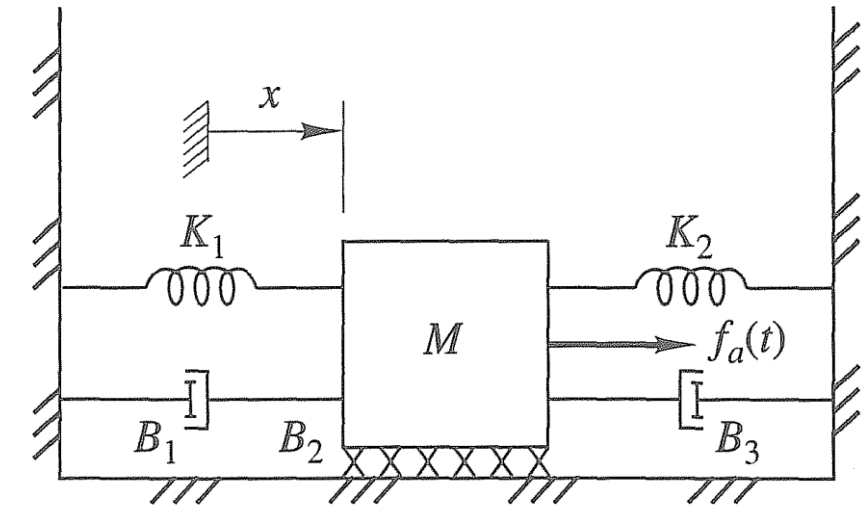
$$f_a(t) = M\ddot{x} + (B_1 + B_2 + B_3)\dot{x} + (K_1 + K_2)x$$

We can see that the two springs and the three damping elements can be replaced by an equivalent element.

$$f_a(t) = M\ddot{x} + B_{eq}\dot{x} + K_{eq}x$$

$$B_{eq} = B_1 + B_2 + B_3$$

$$K_{eq} = K_1 + K_2$$



Free-body Diagram

Modeling of Translational Mechanical Systems

Example 4

When $x_1 = x_2 = 0$, the two springs are neither stretched nor compressed. Draw free-body diagrams for the mass M and for the **massless junction A**, then write the equation of motion of the system. Find K_{eq} for a single spring that could replace the combination of K_1 and K_2 .

Assume that the **positive direction for motion** is to the **left**.

Mass $M \rightarrow$ $f_a(t) - B\dot{x}_1 - K_1(x_1 - x_2) = M\ddot{x}_1$

Junction $A \rightarrow$ $K_1(x_1 - x_2) = K_2x_2$

Solve the second equation for x_2 in terms of x_1 gives:

$$x_2 = \left(\frac{K_1}{K_1 + K_2} \right) x_1 \rightarrow \text{two displacements are proportional to one another}$$

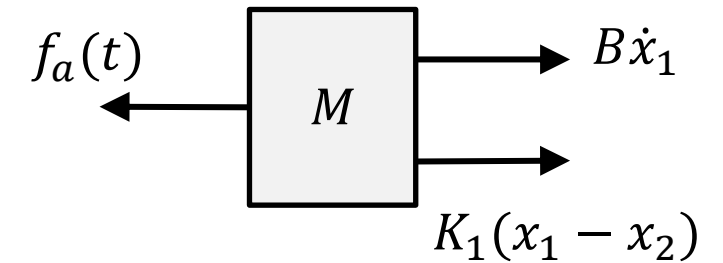
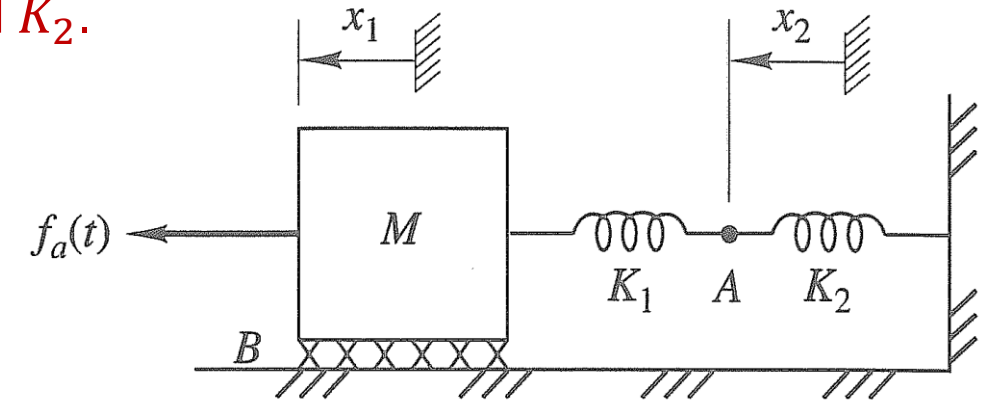
Substituting the x_2 into the first equation we have:

$$f_a(t) - B\dot{x}_1 - K_1 \left(x_1 - \frac{K_1}{K_1 + K_2} x_1 \right) = M\ddot{x}_1 \rightarrow f_a(t) = M\ddot{x}_1 + B\dot{x}_1 + \frac{K_1 K_2}{K_1 + K_2} x_1$$

This equation describes the system formed when the two springs are replaced by a single spring for:

$$K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$$

$$f_a(t) = M\ddot{x}_1 + B\dot{x}_1 + K_{eq}x_1$$



Modeling of Translational Mechanical Systems

Example 5

Draw the free-body diagrams and use Newton's second law to write two modeling equations for the given two-mass system.

The system has **two degrees of freedom**, since each mass can be moved in the horizontal direction while the other is held still.

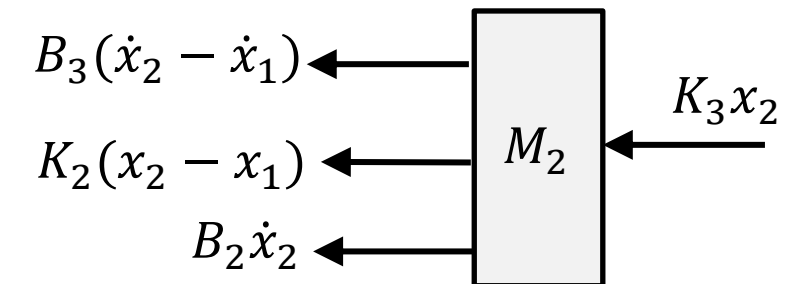
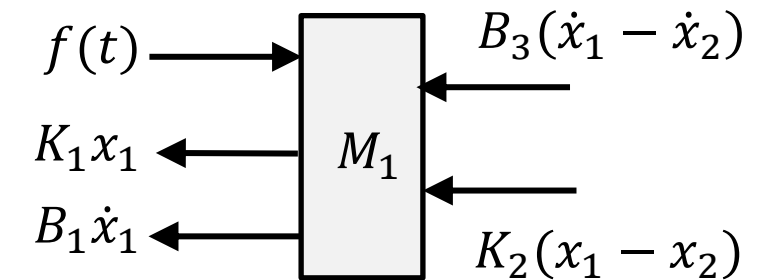
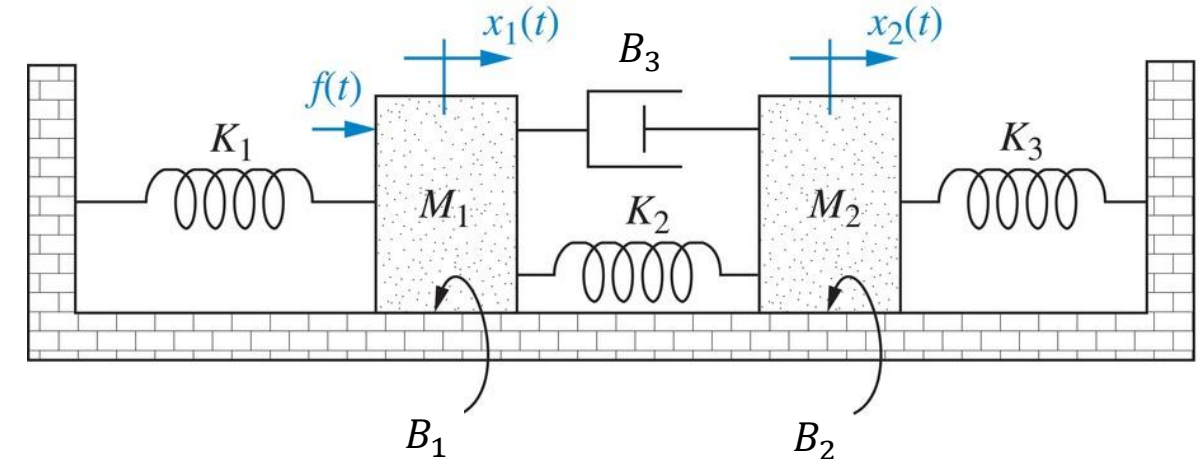
Draw the **free-body diagram** of mass body M_1 and M_2 .

$$\text{Mass } M_1 \rightarrow f(t) - K_1 x_1 - B_1 \dot{x}_1 - B_3(\dot{x}_1 - \dot{x}_2) - K_2(x_1 - x_2) = M_1 \ddot{x}_1$$

$$\text{Mass } M_2 \rightarrow -B_3(\dot{x}_2 - \dot{x}_1) - K_2(x_2 - x_1) - B_2 \dot{x}_2 - K_3 x_2 = M_2 \ddot{x}_2$$

Rearrange the equations to find the **equations of motion**:

$$\begin{cases} M_1 \ddot{x}_1 + (B_1 + B_3) \dot{x}_1 + (K_1 + K_2) x_1 - B_3 \dot{x}_2 - K_2 x_2 = f(t) \\ M_2 \ddot{x}_2 + (B_2 + B_3) \dot{x}_2 + (K_2 + K_3) x_2 - B_3 \dot{x}_1 - K_2 x_1 = 0 \end{cases}$$



Modeling of Translational Mechanical Systems

□ Choosing the Equilibrium Position as Coordinate Reference

- Assume the mass-spring-damper system in **vertical motion**.
- Draw the **free-body diagram** of mass body M including the **gravitational force**.

$$\text{Mass } M \rightarrow f_a(t) + Mg - Kx - B\dot{x} = M\ddot{x} \rightarrow f_a(t) + Mg = M\ddot{x} + B\dot{x} + Kx$$

- Suppose that the applied force $f_a(t)$ is **zero** and the mass is not moving and is in its equilibrium position.

Then $x = x_0$ is the constant displacement caused by the gravitational force.

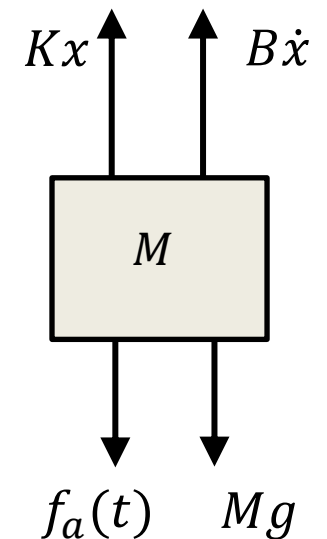
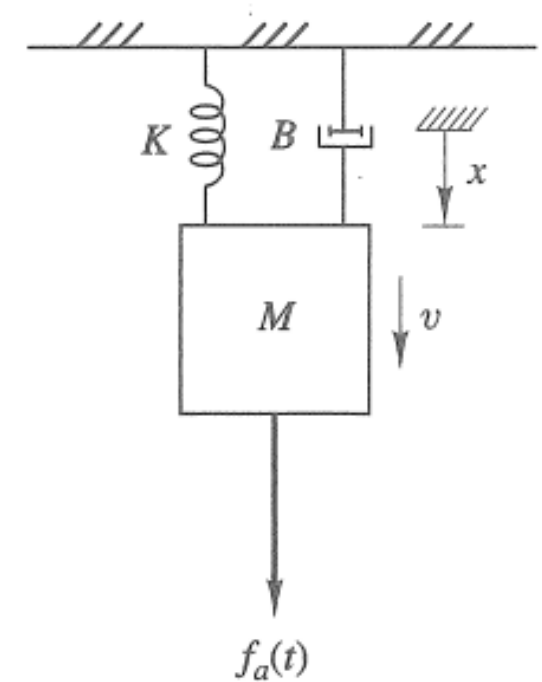
Since, $\dot{x}_0 = \ddot{x}_0 = 0$ the equation of motion is $\rightarrow Mg = Kx_0$

- Now reconsider the case when $f_a(t)$ is **nonzero** and the mass is moving from its equilibrium position.

Then $x = x_0 + z$, where z is the additional displacement caused by $f_a(t)$

$$f_a(t) + Mg = M\ddot{z} + B\dot{z} + K(x_0 + z) \rightarrow f_a(t) = M\ddot{z} + B\dot{z} + Kz$$

- For a mass connected to a spring element, the force due to gravity Mg is canceled out of the equation of motion by the force in the spring due to its **static deflection**, **as long the displacement of the mass is measured from the equilibrium position**.
- This means that if we are defined the vertical displacement from the static-equilibrium position caused by the gravitational force, then **no need to show the Mg force** in our free-body diagram.



Modeling of Translational Mechanical Systems

□ Displacement Input

Example 6

The suspension system for one wheel of an old-fashioned pickup truck is shown in the figure.

The mass of the vehicle is m_1 and the mass of the wheel is m_2 .

The suspension spring has a spring constant k_1 and the tire has a spring constant k_2 .

The damping constant of the shock absorber is B .

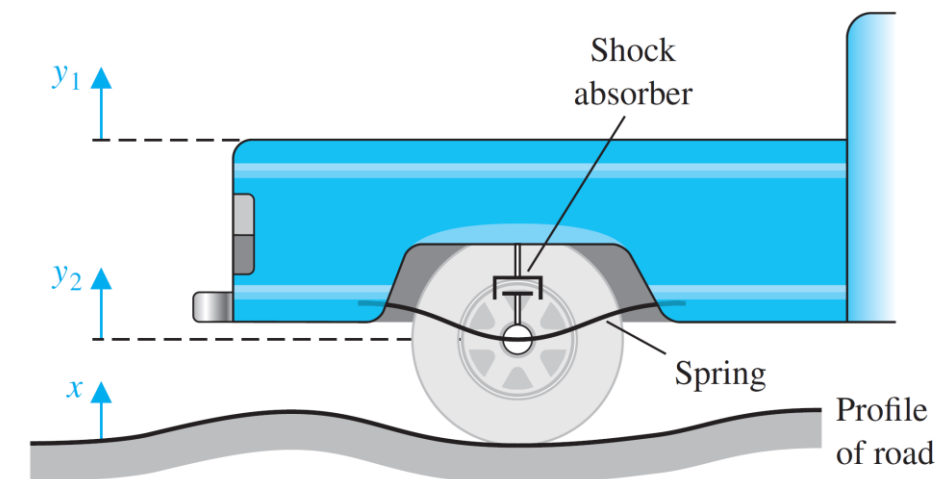
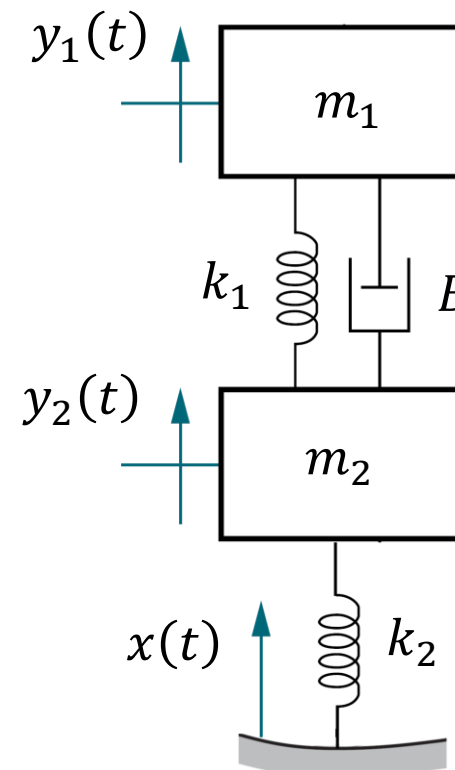
Derive the equations of motion for m_1 and m_2 in terms of the displacements from equilibrium, y_1 and y_2 with $x(t)$ as the input.

Assume that the mass is traveling upward, the **suspension system** can be modeled as a following **mass-spring-damper system**.

The system has **two degrees of freedom**, since each mass can be moved in the vertical direction while the other is held still.

Draw the **free-body diagram** of the system for each mass.

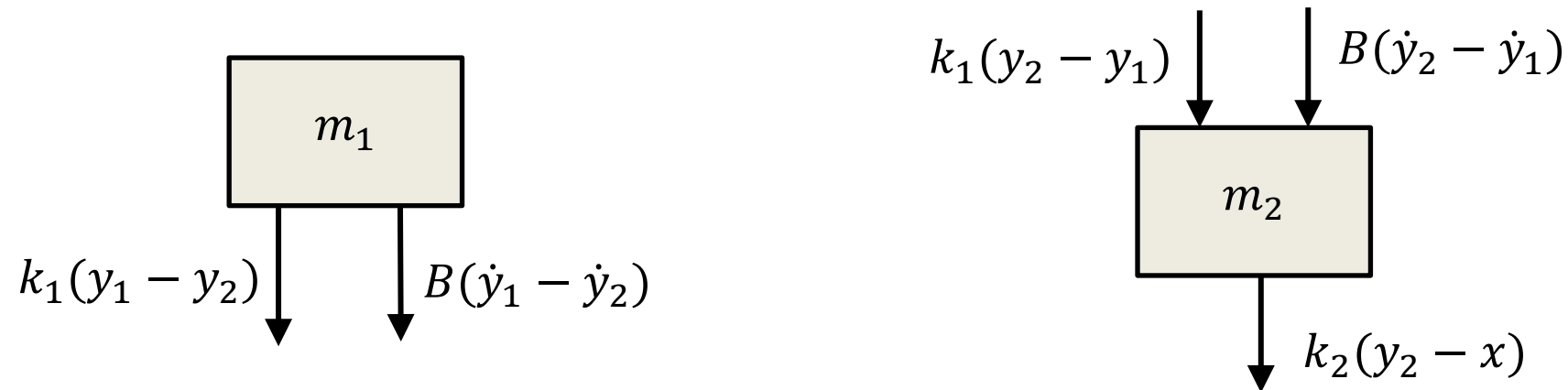
Place all the forces felt by the mass.



Modeling of Translational Mechanical Systems

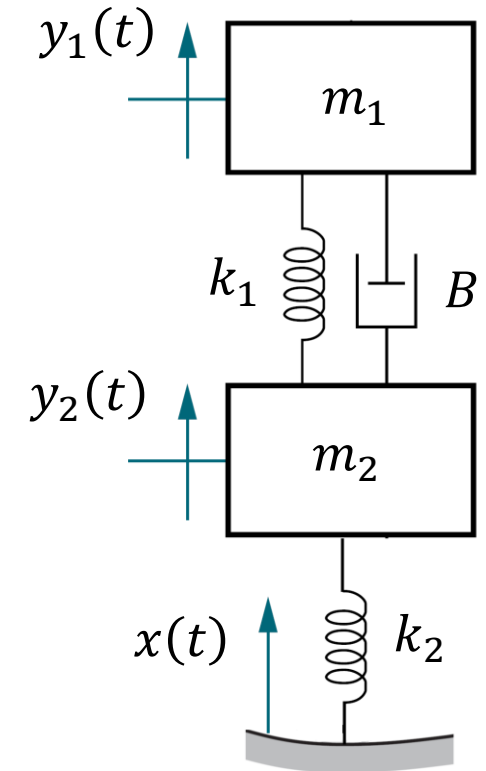
Example 6

Draw the free-body diagram of mass body m_1 and m_2 .



For mass $m_1 \rightarrow -k_1(y_1 - y_2) - B(\dot{y}_1 - \dot{y}_2) = m_1\ddot{y}_1$ **Eqn.1**

For mass $m_2 \rightarrow -k_1(y_2 - y_1) - B(\dot{y}_2 - \dot{y}_1) - k_2(y_2 - x) = m_2\ddot{y}_2$ **Eqn. 2**



Note that since the **gravitational force** is canceled out by the **static spring force**, no need to include gravity in the equations.

Rearrange the equations to find the **equations of motion**:

$$\begin{cases} m_1\ddot{y}_1 + B\dot{y}_1 + k_1y_1 - B\dot{y}_2 - k_1y_2 = 0 \\ m_2\ddot{y}_2 + B\dot{y}_2 + (k_1 + k_2)y_2 - B\dot{y}_1 + k_1y_1 - k_2x = 0 \end{cases}$$

THANK YOU